Friday 9:27 PM

Give me 100 questions of Excel with respective formula for solution using same dataset for a professional Data Analyst.

Absolutely. Below is a **100-question Excel practice set for a professional Data Analyst**, all designed around **one consistent dataset** so you can progressively practice data cleaning, analysis, lookup, aggregation, dates, text, dynamic arrays, statistics, and dashboard-oriented calculations.

Dataset: `SalesData`

Use these columns:

Column	Field
A	Order_ID
B	Order_Date
C	Customer_ID
D	Customer_Name
E	Gender
F	Age
G	Region
H	Country
I	Product_ID
J	Product_Name

Column	Field
K	Category
L	Subcategory
M	Salesperson
N	Quantity
O	Unit_Price
P	Discount
Q	Sales
R	Cost
S	Profit
T	Payment_Method
U	Order_Status
V	Customer_Rating
W	Delivery_Date

Convert the range to an Excel Table named `SalesData`.

Assume:

- `Sales = Quantity × Unit_Price × (1 - Discount)`
- `Profit = Sales - Cost`
- `Discount` is stored as a decimal, e.g. `10% = 0.10`.

Part 1 — Basic Data Analysis

1. Calculate total sales.

```excel
=SUM(SalesData[Sales])
```

2. Calculate total profit.

```excel
=SUM(SalesData[Profit])
```

3. Calculate total quantity sold.

```excel
=SUM(SalesData[Quantity])
```

4. Calculate average sales per order.

```excel
=AVERAGE(SalesData[Sales])
```

5. Calculate average profit per order.

```excel
=AVERAGE(SalesData[Profit])
```

6. Find the maximum sales value.

```excel
=MAX(SalesData[Sales])
```

7. Find the minimum sales value.

```excel
=MIN(SalesData[Sales])
```

8. Find the maximum profit.

```excel
=MAX(SalesData[Profit])
```

9. Find the average customer rating.

excel

```
=AVERAGE(SalesData[Customer_Rating])
```

10. Count total orders.

excel

```
=COUNTA(SalesData[Order_ID])
```

Part 2 — Counting and Conditional Analysis

11. Count orders from the North region.

excel

```
=COUNTIF(SalesData[Region],"North")
```

12. Count orders from the South region.

excel

```
=COUNTIF(SalesData[Region],"South")
```

13. Count orders where sales exceed 10,000.

excel

```
=COUNTIF(SalesData[Sales],">10000")
```

14. Count orders with profit greater than 2,000.

excel

```
=COUNTIF(SalesData[Profit],">2000")
```

15. Count customers aged above 40.

excel

```
=COUNTIF(SalesData[Age],">40")
```

16. Count female customers.

```excel
=COUNTIF(SalesData[Gender],"Female")
```

17. Count completed orders.

```excel
=COUNTIF(SalesData[Order_Status],"Completed")
```

18. Count cancelled orders.

```excel
=COUNTIF(SalesData[Order_Status],"Cancelled")
```

19. Count orders paid by Credit Card.

```excel
=COUNTIF(SalesData[Payment_Method],"Credit Card")
```

20. Count orders with rating of 5.

```excel
=COUNTIF(SalesData[Customer_Rating],5)
```

Part 3 — SUMIF / SUMIFS

21. Calculate sales for the North region.

```excel
=SUMIF(SalesData[Region],"North",SalesData[Sales])
```

22. Calculate profit for the North region.

```excel
```

```
=SUMIF(SalesData[Region],"North",SalesData[Profit])
```

23. Calculate sales for the Electronics category.

excel

```
=SUMIF(SalesData[Category],"Electronics",SalesData[Sales])
```

24. Calculate profit for Electronics.

excel

```
=SUMIF(SalesData[Category],"Electronics",SalesData[Profit])
```

25. Calculate sales generated by a specific salesperson.

excel

```
=SUMIF(SalesData[Salesperson],"John",SalesData[Sales])
```

26. Calculate sales for Electronics in the North region.

excel

```
=SUMIFS(
SalesData[Sales],
SalesData[Category],"Electronics",
SalesData[Region],"North"
)
```

27. Calculate profit from completed orders.

excel

```
=SUMIF(
SalesData[Order_Status],
"Completed",
SalesData[Profit]
)
```

28. Calculate sales for customers aged above 40.

excel

```
=SUMIF(SalesData[Age],">40",SalesData[Sales])
```

29. Calculate sales from customers aged between 25 and 40.

```excel
=SUMIFS(
SalesData[Sales],
SalesData[Age],">=25",
SalesData[Age],"<=40"
)
```

30. Calculate sales from the North region for Electronics.

```excel
=SUMIFS(
SalesData[Sales],
SalesData[Region],"North",
SalesData[Category],"Electronics"
)
```

Part 4 — Date Analysis

31. Extract the year from Order_Date.

```excel
=YEAR([@Order_Date])
```

32. Extract the month number.

```excel
=MONTH([@Order_Date])
```

33. Extract the month name.

```excel
=TEXT([@Order_Date],"mmmm")
```

34. Extract the quarter.

excel

```
="Q"&ROUNDUP(MONTH([@Order_Date])/3,0)
```

35. Extract the weekday name.

excel

```
=TEXT([@Order_Date],"dddd")
```

36. Calculate delivery duration in days.

excel

```
=[@Delivery_Date]-[@Order_Date]
```

37. Calculate average delivery time.

excel

```
=AVERAGE(SalesData[Delivery_Date]-SalesData[Order_Date])
```

For older Excel versions, use a helper column first.

38. Count orders placed in 2025.

excel

```
=COUNTIFS(
SalesData[Order_Date],">="&DATE(2025,1,1),
SalesData[Order_Date],"<"&DATE(2026,1,1)
)
```

39. Calculate sales for 2025.

excel

```
=SUMIFS(
SalesData[Sales],
SalesData[Order_Date],">="&DATE(2025,1,1),
SalesData[Order_Date],"<"&DATE(2026,1,1)
)
```

40. Calculate sales for January 2025.

excel

```
=SUMIFS(
SalesData[Sales],
SalesData[Order_Date],">="&DATE(2025,1,1),
SalesData[Order_Date],"<"&DATE(2025,2,1)
)
```

Part 5 — Profitability Analysis

41. Calculate profit margin for each order.

excel

```
=IFERROR([@Profit]/[@Sales],0)
```

42. Calculate overall profit margin.

excel

```
=SUM(SalesData[Profit])/SUM(SalesData[Sales])
```

43. Calculate sales after discount.

excel

```
=[@Quantity]*[@Unit_Price]*(1-[@Discount])
```

44. Calculate gross profit.

excel

```
=[@Sales]-[@Cost]
```

45. Identify profitable vs. loss-making orders.

excel

```
=IF([@Profit]>0,"Profitable","Loss")
```

46. Identify high-profit orders.

excel

```
=IF([@Profit]>=2000,"High Profit","Normal")
```

47. Calculate profit per unit.

excel

```
=IFERROR([@Profit]/[@Quantity],0)
```

48. Calculate average order value.

excel

```
=AVERAGE(SalesData[Sales])
```

49. Calculate revenue per customer.

excel

```
=SUM(SalesData[Sales])/COUNTA(UNIQUE(SalesData[Customer_ID]))
```

50. Calculate profit as a percentage of cost.

excel

```
=SUM(SalesData[Profit])/SUM(SalesData[Cost])
```

Part 6 — IF / IFS / Logical Analysis

51. Categorize customers by age.

excel

```
=IFS(
[@Age]<25,"Young",
[@Age]<40,"Adult",
[@Age]<60,"Middle Age",
TRUE,"Senior"
)
```

52. Categorize sales.

excel

```
=IFS(
[@Sales]<1000,"Low",
[@Sales]<5000,"Medium",
TRUE,"High"
)
```

53. Categorize profit margin.

excel

```
=IFS(
[@Profit]/[@Sales]<0.10,"Low Margin",
[@Profit]/[@Sales]<0.25,"Medium Margin",
TRUE,"High Margin"
)
```

54. Identify delayed deliveries.

excel

```
=IF(
[@Delivery_Date]-[@Order_Date]>7,
"Delayed",
"On Time"
)
```

55. Identify high-value customers.

excel

```
=IF([@Sales]>=10000,"High Value","Regular")
```

56. Identify discounted orders.

excel

```
=IF([@Discount]>0,"Discounted","No Discount")
```

57. Identify orders requiring attention.

excel

```
=IF(
OR([@Profit]<0,[@Order_Status]="Cancelled"),
```

```
"Attention Required",
"Normal"
)
```

58. Identify successful orders.

excel

```
=IF(
AND([@Profit]>0,[@Order_Status]="Completed"),
"Successful",
"Other"
)
```

59. Identify customers with high ratings.

excel

```
=IF([@Customer_Rating]>=4,"Satisfied","Needs Attention")
```

60. Create a sales performance category.

excel

```
=IFS(
AND([@Sales]>=10000,[@Profit]>2000),"Excellent",
[@Sales]>=5000,"Good",
TRUE,"Low"
)
```

Part 7 — Lookup Functions

Assume you create a separate table called `ProductData`:

Product_ID	Product_Name	Category	Unit_Cost

61. Find Product Name from Product_ID using XLOOKUP.

excel

```
=XLOOKUP([@Product_ID],ProductData[Product_ID],ProductData[Product_Name])
```

62. Find Category.

excel

```
=XLOOKUP(
[@Product_ID],
ProductData[Product_ID],
ProductData[Category]
)
```

63. Find Unit Cost.

excel

```
=XLOOKUP(
[@Product_ID],
ProductData[Product_ID],
ProductData[Unit_Cost]
)
```

64. Use XLOOKUP with a default message.

excel

```
=XLOOKUP(
[@Product_ID],
ProductData[Product_ID],
ProductData[Product_Name],
"Not Found"
)
```

65. Use VLOOKUP to find Product Name.

excel

```
=VLOOKUP(
[@Product_ID],
ProductData[[Product_ID]:[Product_Name]],
2,
FALSE
)
```

66. Use INDEX/MATCH to find Category.

excel

```
=INDEX(
ProductData[Category],
```

```
MATCH([@Product_ID],ProductData[Product_ID],0)
```

67. Find the salesperson for a particular Order_ID.

excel

```
=XLOOKUP(
A2,
SalesData[Order_ID],
SalesData[Salesperson]
)
```

68. Find customer name from Customer_ID.

excel

```
=XLOOKUP(
C2,
SalesData[Customer_ID],
SalesData[Customer_Name]
)
```

69. Find sales for a specific Order_ID.

excel

```
=XLOOKUP(
A2,
SalesData[Order_ID],
SalesData[Sales]
)
```

70. Find profit for a specific Order_ID.

excel

```
=XLOOKUP(
A2,
SalesData[Order_ID],
SalesData[Profit]
)
```

Part 8 — Text Analysis

71. Extract the first name from Customer_Name.

excel

```
=TEXTBEFORE([@Customer_Name]," ")
```

72. Extract the last name.

excel

```
=TEXTAFTER([@Customer_Name]," ")
```

73. Convert customer names to uppercase.

excel

```
=UPPER([@Customer_Name])
```

74. Convert customer names to lowercase.

excel

```
=LOWER([@Customer_Name])
```

75. Proper-case customer names.

excel

```
=PROPER([@Customer_Name])
```

76. Count characters in customer names.

excel

```
=LEN([@Customer_Name])
```

77. Remove unnecessary spaces.

excel

```
=TRIM([@Customer_Name])
```

78. Check whether product name contains "Laptop".

excel

```
=IF(ISNUMBER(SEARCH("Laptop",[@Product_Name])),"Laptop","Other")
```

79. Combine Customer_Name and Country.

```excel
=[@Customer_Name]&" - "&[@Country]
```

80. Create a customer-region identifier.

```excel
=[@Customer_ID]&"-"&[@Region]
```

Part 9 — Ranking and Top-N Analysis

81. Rank orders by sales.

```excel
=RANK.EQ([@Sales],SalesData[Sales],0)
```

82. Rank orders by profit.

```excel
=RANK.EQ([@Profit],SalesData[Profit],0)
```

83. Find the highest sales value.

```excel
=LARGE(SalesData[Sales],1)
```

84. Find the fifth-highest sales.

```excel
=LARGE(SalesData[Sales],5)
```

85. Find the lowest sales value.

excel

```
=SMALL(SalesData[Sales],1)
```

86. Find the top 10 sales values.

excel

```
=LARGE(SalesData[Sales],SEQUENCE(10))
```

87. Return the top 10 orders.

excel

```
=TAKE(
SORT(SalesData,SalesData[Sales],-1),
10
)
```

88. Return the top 10 customers by total sales.

excel

```
=LET(
u,UNIQUE(SalesData[Customer_ID]),
s,SUMIF(SalesData[Customer_ID],u,SalesData[Sales]),
TAKE(SORT(HSTACK(u,s),2,-1),10)
)
```

89. Return the top 5 products by sales.

excel

```
=LET(
u,UNIQUE(SalesData[Product_Name]),
s,SUMIF(SalesData[Product_Name],u,SalesData[Sales]),
TAKE(SORT(HSTACK(u,s),2,-1),5)
)
```

90. Return the bottom 5 products by sales.

excel

```
=LET(
u,UNIQUE(SalesData[Product_Name]),
s,SUMIF(SalesData[Product_Name],u,SalesData[Sales]),
```

```
TAKE(SORT(HSTACK(u,s),2,1),5)
)
```

Part 10 — Advanced Data Analyst Questions

91. Calculate each order's percentage contribution to total sales.

excel

```
=[@Sales]/SUM(SalesData[Sales])
```

92. Calculate each product's contribution to total profit.

excel

```
=SUMIF(
SalesData[Product_Name],
[@Product_Name],
SalesData[Profit]
)/SUM(SalesData[Profit])
```

93. Calculate cumulative sales.

Assuming data is sorted by date:

excel

```
=SUM($Q$2:Q2)
```

94. Calculate cumulative profit.

excel

```
=SUM($S$2:S2)
```

95. Calculate Year-to-Date sales.

excel

```
=SUMIFS(
SalesData[Sales],
SalesData[Order_Date],">="&DATE(YEAR(B2),1,1),
SalesData[Order_Date],"<="&B2
)
```

96. Calculate Month-to-Date sales.

```excel
=SUMIFS(
SalesData[Sales],
SalesData[Order_Date],">="&EOMONTH(B2,-1)+1,
SalesData[Order_Date],"<="&B2
)
```

97. Calculate YoY sales growth.

Assume `B2` contains current-year sales and `C2` contains previous-year sales.

```excel
=IFERROR((B2-C2)/C2,0)
```

Or:

```excel
=IFERROR(B2/C2-1,0)
```

98. Calculate repeat-customer percentage.

```excel
=LET(
u,UNIQUE(SalesData[Customer_ID]),
orders,COUNTIF(SalesData[Customer_ID],u),
COUNTIF(orders,">1")/COUNTA(u)
)
```

99. Calculate the median order value.

```excel
=MEDIAN(SalesData[Sales])
```

100. Identify sales outliers using the IQR method.

Q1:

```excel
```

```
=QUARTILE.INC(SalesData[Sales],1)
```

Q3:

```excel
=QUARTILE.INC(SalesData[Sales],3)
```

IQR:

```excel
=Q3-Q1
```

Lower Bound:

```excel
=Q1-1.5*IQR
```

Upper Bound:

```excel
=Q3+1.5*IQR
```

Identify each outlier:

```excel
=IF(
OR(
[@Sales]<Lower_Bound,
[@Sales]>Upper_Bound
),
"Outlier",
"Normal"
)
```

Professional Data Analyst Skills Covered

These 100 questions progressively cover the Excel skills you would actually use in data-analysis work:

Skill	Questions
Basic aggregation	1–10
COUNTIF / COUNTIFS	11–20
SUMIF / SUMIFS	21–30
Date analysis	31–40
Profitability	41–50
IF / IFS / logical functions	51–60
XLOOKUP / VLOOKUP / INDEX-MATCH	61–70
Text functions	71–80
Ranking / Top-N	81–90
Advanced analytics	91–100

For professional-level practice, don't just enter the formulas. Try to build an Excel workbook with these sheets:

1. **Raw_Data** — original dataset
2. **Cleaned_Data** — cleaned/validated data
3. **Analysis** — questions 1–100
4. **Pivot_Analysis** — PivotTables
5. **Dashboard** — KPIs, trends, Top 10 products/customers, regional performance, profitability, and slicers

This gives you a single dataset that can be used to practice **Excel formulas → data cleaning → analytical calculations → PivotTables → dashboard development**, which is much closer to a real Data Analyst workflow than practicing isolated formulas.

Monday 11:10 AM

how to calculate age to till date from birthday?

If you have a **Date of Birth** in Excel and want to calculate the person's **current age as of today**, use `DATEDIF`.

1. Age in completed years

Suppose the date of birth is in **A2**:

```excel
=DATEDIF(A2,TODAY(),"Y")
```

For example:

Date of Birth	Formula	Age
15-Mar-1995	`=DATEDIF(A2,TODAY(),"Y")`	31

This returns the person's **completed age in years**.

2. Age in years + months + days

If you want something like **31 years, 6 months, 20 days**:

```excel
=DATEDIF(A2,TODAY(),"Y")&" Years, "&DATEDIF(A2,TODAY(),"YM")&" Months, "&DATEDIF(A2,TODAY
```

3. Age in total months

```excel
=DATEDIF(A2,TODAY(),"M")
```

4. Age in total days

```excel
=DATEDIF(A2,TODAY(),"D")
```

5. If you want age as of a specific date

For example, calculate age as of **31-Dec-2026**:

```excel
=DATEDIF(A2,DATE(2026,12,31),"Y")
```

For your Data Analyst dataset

If your `SalesData` table has a `Date_of_Birth` column, add an **Age** column:

excel

```
=DATEDIF([@Date_of_Birth],TODAY(),"Y")
```

This is better than simply using:

excel

```
=YEAR(TODAY())-YEAR(A2)
```

because the latter can give an incorrect age when the person's birthday **hasn't occurred yet this year**.

ChatGPT can make mistakes. Check important info.